

Link

- <http://classwork.engr.oregonstate.edu:43453/>

Group Review 1

Hello!

The overview effectively communicates that Seasick Steve's Guitar Emporium is growing larger and needs a better management system in the form of a database to keep track of inventory and sales details. I like that you list the specific locations of the storefronts. It may be good to add some placeholder metrics alongside it (like an average net profit of \$100,000 annually) to show the scale of the company.

Over four entities are described: Sales, Employees, Customers, Guitars, and Purchases; each representing a single idea that can be stored as a list. The naming convention is consistent throughout the entities, with each table name being capitalized and plural and the attributes being lowercase and singular.

The entity details fully describe their purposes, and attributes are listed with their data types and constraints. Something I noticed is that, depending on your implementation plan, you may want to add a unique constraint to email and/or phone number to ensure that contact details are only associated with one customer account. Additionally, you have noted the relationships between the entities well

Your 1:M relationships look to be correctly formulated, though no M:N relationships have been described. There is also a mismatch between the varchar allowed characters in the outline and the varchar allowed characters in the EDR.

Good job!

I have authored this feedback without outside sources beyond the assignment description.

Group Review 2

Hi Thomas / Nolan,

The overview describes the problem to be solved which is primarily about buying and selling new or used guitars and requiring an upgrade to the backend in order to support scalability and growth. The need surrounds inventory management to track sales and purchases along with accurate inventory.

The overview does not list specific facts. Some facts that could help better support the scope would be metrics related to how much the store has grown either by rate, revenue sales, or volume sales. That would help illustrate the scale of growth the business has gained traction for the need to show why their current system doesn't support the complexity of their operations anymore. This would also help provide more context to customer growth.

There are at least 4 entities described. The entities include: sales, employees, customers, guitars, and purchases. Sales calculates customer sales for all products i.e. sold items by customers and location. Employees include the employees that work at a given shop. Customers track all of the customers who have made purchases. Guitars are the inventory per store. Purchases include keeping track of all purchases per customer per location. It may help to rename Guitars to Inventory so that it is more explicit that you're tracking inventory there. This would also help with scalability of the database if you later decide to include other equipment or accessories as part of the inventory list.

The outline details the purpose, attribute datatypes, and relationships between entities. The sales relationships include M:1 to customers, employees and guitars. Employee relationships include 1:M relationships with sales and purchases. Customer relationships include 1:M to sales and purchases. Guitars relationships include 1:M with sales and purchases. Purchases relationships include M:1 with customers, employees, and guitars. Purchases and sales could potentially be consolidated. Otherwise, they may need a relationship. For employees, if sales are commissions based it, may be useful to include an attribute for commissions based on the sales and purchases.

There are 1:M relationships. There are no M:M relationships. The ERD presents a logical view of the database.

Purchases could potentially be consolidated with Sales. It's not that clear on why they are separate entities. They share similar attributes. In current state, Sales and

Purchases also do not have a relationship but perhaps they should if they are kept separate.

- There is consistency in the naming between overview and entity / attributes.
- All entities are plural except the Customer table. The outline says “Customers” while the ERD says “Customer”.
- Capitalization for naming the entities is consistent. Lowercase attributes are also consistent.

All my work in entirety was original with no other assistance.

Group Review 3

Hi Thomas & Nolan,

I see through your overview that you'd like to make a software to manage the data & operations of an expanding guitar sale business. The business is steadily growing and will definitely need a proper system to manage that data, as well as be able to analyze sales tracking metrics to further grow, rather than manual input as you said. Even though for a group of 2, I see you've picked 5 different entities being Sales, Employees, Customers, Guitars, and Purchases. Each of which has a fully described purpose, reasonable attributes & datatypes, and relationships. To me, the relationships all look correctly formatted, except for one issue. I don't see any M:N relationships, which at least 1 is needed. The ERD also shows a logical & direct view of the database, and I was able to immediately understand it at a glance. As for consistency in naming, I did notice that all the tables were plural except for the "Customer" table, which sticks out from the rest of the naming.

This review was given entirely with my own words based on what I saw in the project document, and with no other assistance.

Actions Based on Feedback

Based on the above feedback, I added a “Lessons” entity in order to create an M:N relationship in my database, and added attributes to Employees in order to reflect which employees are teachers, and what instruments they can teach. I added a price attribute to the guitar entity to make the prices for each guitar clearer as well.

I've gotten rid of the purchases table as it really wasn't necessary, and made working on the project a lot more difficult. I added HTML/CSS for the UI, as well as data manipulation queries for the backend.

Seasick Steve's Guitar Emporium

Team Members

- Thomas Munro, Nolan Flores

Overview

- Seasick Steve's Guitar Emporium is struggling to maintain their growth as their store gains more traction. This business buys and sells new or used guitars & basses to musicians with two locations, one in Astoria, OR, and another in Newport, OR. The store also provides guitar, bass, piano, drum, and vocal lessons for students at a rate of \$25 for half an hour, or \$50 for an hour. They've been steadily growing their business over the past few years, and have gotten to a point where they need to upgrade their backend in order to support new customers and further growth for the business. More customers means a need for better management of inventory for the products being sold, and better ways to keep track of each sale or purchase that the shop makes. The idea is to streamline the flow of commerce and ensure that customers will get an accurate idea of the gear that they're buying.

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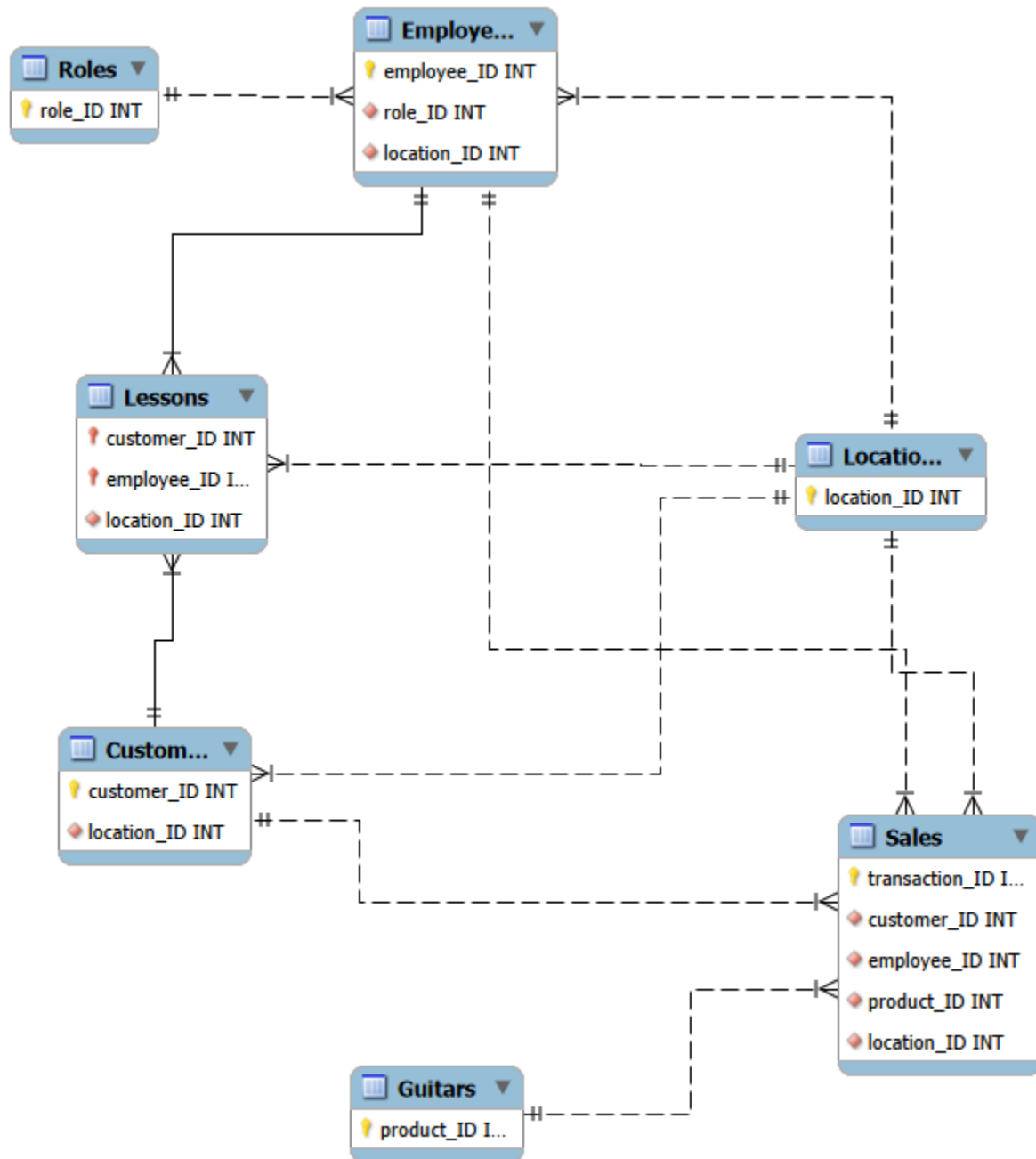
Database Outline

- Sales
 - Tabulates all customer sales for all products, including the guitar that was sold, the customer, the employee who sold the items, and the location where the item was sold.
 - transaction_ID: INT, auto_increment, unique, not NULL, PK
 - customer_ID: INT, not NULL, FK
 - employee_ID: INT, not NULL, FK
 - product_ID: INT, not NULL, FK
 - s_date: DATE, not NULL
 - amount: FLOAT, not NULL, format of 00.00
 - location_ID: INT, not NULL, FK
 - Relationships
 - M:1 relationship from Sales to Customers
 - M:1 relationship from Sales to Employees
 - M:1 relationship with Guitars
 - M:1 with Locations
- Locations
 - Stores all store locations, along with each store's address and phone number.
 - location_ID: INT, not NULL, unique
 - l_address: varchar(45), not NULL
 - phone_number: varchar(45), not NULL
 - Relationships
 - 1:M relationship with Employees
 - 1:M relationship with Sales
 - 1:M relationship with Lessons
 - 1:M relationship with Purchases
- Employees
 - This entity provides information for each employee that works at the shop. This is to keep track of their pay, their role, the location they work, and keep track of who makes sales.
 - employee_ID: INT, auto_increment, unique, not NULL, PK
 - e_name: varchar(45), not NULL
 - role_ID: INT, not NULL, FK
 - location_ID: INT, not NULL, FK

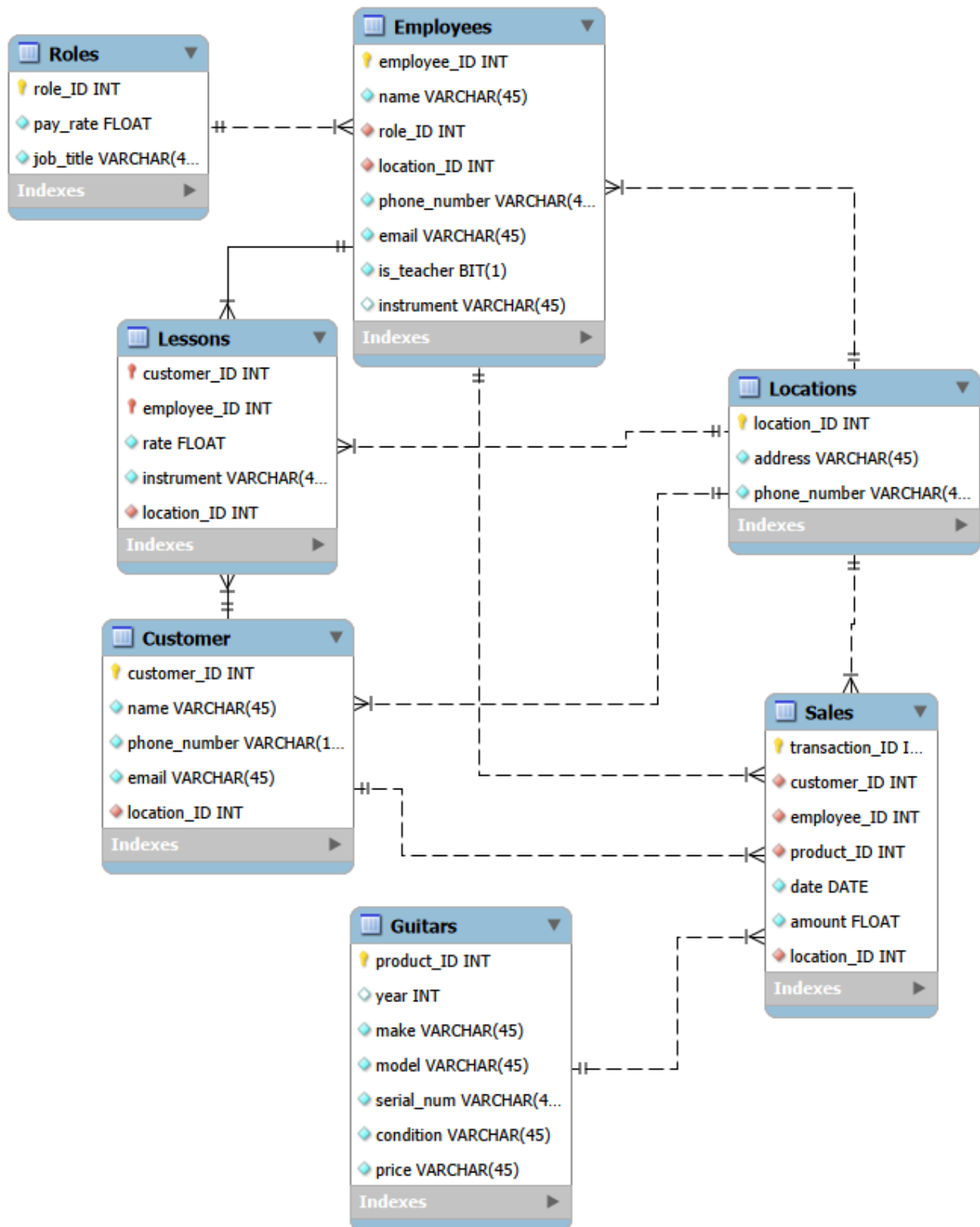
- phone_number: varchar(45), not NULL
 - email: varchar(45), not NULL
 - is_Teacher: BIT(1), not NULL
 - instrument: varchar(45)
- Relationships
 - 1:M relationship with Sales
 - 1:M relationship with Purchases
 - M:1 relationship with Locations
 - M:N relationship with Customers through Lessons
- Roles
 - This entity keeps track of the different job titles that employees will have, along with their rate of pay.
 - role_ID: INT, not NULL, unique
 - pay_rate: FLOAT, not NULL, format of 00.00
 - job_title: varchar(45)
 - Relationships
 - 1:M relationship with Employees
- Customers
 - This entity keeps track of all customers who do business with the company.
 - customer_ID: INT, auto_increment, unique, not NULL, PK
 - c_name: varchar(45), not NULL
 - phone_number: varchar(16), not NULL
 - email: varchar(45), not NULL
 - Relationships
 - 1:M relationship with Sales
 - 1:M relationship with Purchases
 - M:N relationship with Employees through Lessons
- Guitars
 - This entity keeps track of all guitars & basses in the inventory, and stores information about the make, model, year, condition, and serial number (a unique number printed onto a guitar, different from the product number the store gives).
 - product_ID: INT, auto_increment, unique, not NULL, PK
 - year: INT, not NULL
 - brand: varchar(45), not NULL
 - model: varchar(45), not NULL

- serial_num: varchar(45), not NULL
 - condition: varchar(45), not NULL
 - amount: float, not NULL, format of 00.00
- Relationships
 - 1:M relationship with Sales
 - 1:M relationship with Purchases
- Lessons
 - This entity keeps track of which customers take lessons with which employee, along with the rate, the instrument being taught, and the location of the lessons. Customers can take lessons with multiple employees, and employees will have multiple customers as their students.
 - customer_ID: INT, not NULL, FK
 - employee_ID: INT, not NULL, FK
 - rate: FLOAT, not NULL, format of 00.00
 - instrument: varchar(45), NOT NULL
 - location_ID: INT, not NULL, FK
 - Relationships
 - 1:M relationship with Customers
 - 1:M relationship with Employees
 - M:1 relationship with Locations

ERD Diagram



Schema Diagram



Tables

Employees								Locations			Roles		
employee_ID	name	role_ID	location_ID	phone_num	email	is_teacher	instrument	location_ID	address	phone_number	role_ID	pay_rate	job_title
1	James Jamerson		1	2	9713292104	email1@gmail.com	0 NULL	1	Newport, OR	9713367324	1	\$16	Sales
2	Jim Johnman		1	1	9713292105	email2@gmail.com	0 NULL	2	Astoria, OR	9713292108	2	\$25	Teacher
3	John Jimman		2	2	9713292106	email3@gmail.com	1 Guitar/Bass				3	\$30	Manager
4	Man Jimjohn		1	1	9713292108	email4@gmail.com	0 NULL						
5	Steve Screeb		3	2	9713367324	email5@gmail.com	0 NULL						
6	Carol Crumb		3	1	9713343532	email6@gmail.com	0 NULL						
Customers													
Customer_ID	name	phone_number	email	is_Student									
1	John Smith	9713282104	email6@gmail.com	1									
2	Smith John	9723487283	email5@gmail.com	1									
3	Cruella Deville	9713282105	email2@gmail.com	1									
Guitars													
product_ID	year	make	model	condition	price								
1	1973	Fender	Precision Bass	Good	\$3,000								
2	2009	Epiphone	Riviera	Great	\$900								
3	1962	Fender	Telecaster	Good	\$5,000								
4	2022	Fender	Player Jazz Bass	Good	\$600								
5	2013	Gibson	Les Paul	Mint	\$3,000								
Sales													
transaction_ID	customer_ID	employee_ID	product_ID	location_ID	date	amount							
1	1	1	2	3	1	3/12/2024	\$3,000						
2		2	2	4	2	4/9/2024	\$900						
3		3	4	5	2	6/13/2024	\$5,000						
Lessons													
customer_ID	employee_ID	rate	instrument	location_ID									
1		3	\$25 Guitar	2									
2		3	\$25 Guitar	2									
3		3	\$25 Bass	2									
Purchases													
transaction_ID	customer_ID	employee_ID	product_ID	location_ID	date	amount							
1		1	2	3	1	3/12/2024	\$3,000						
2		2	2	4	2	4/9/2024	\$900						
3		3	4	5	2	6/13/2024	\$5,000						

Citations

None.